

Computer Networking Notes

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About This Notes

This document is part of my personal Computer Networking learning journey. These notes are organized to help me build a strong foundation in networking through theory, practical labs, and real-world examples. They are intended for educational purposes and may be updated as my knowledge grows.

Version: 1.0

Introduction to Computer Networks

What is a Computer Network?

A **computer network** is a group of two or more devices connected together so they can communicate and share resources.

These devices can include:

- Computers
- Laptops
- Smartphones
- Servers
- Printers
- Routers
- Switches
- IP Cameras
- Access Points

The connection can be through:

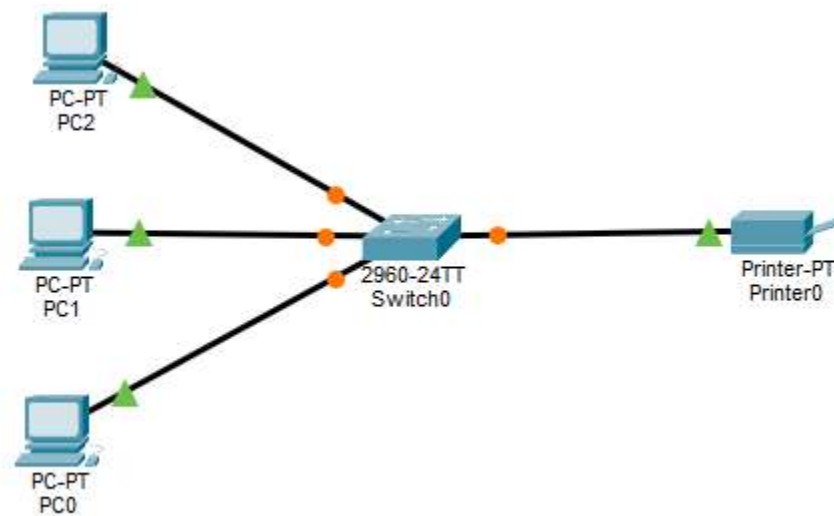
- Ethernet Cable (Wired)
 - Wi-Fi (Wireless)
 - Fiber Optic Cable
 - Cellular Network (4G/5G)
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Real-Life Example

Imagine you have:

- 3 Computers
- 1 Printer
- 1 Switch

All devices are connected using Ethernet cables.



Now:

- PC1 can print documents.
- PC2 can send files to PC3.
- All devices can communicate with each other.

This is called a **Local Area Network (LAN)**.

Why Do We Need Computer Networks?

Without networking:

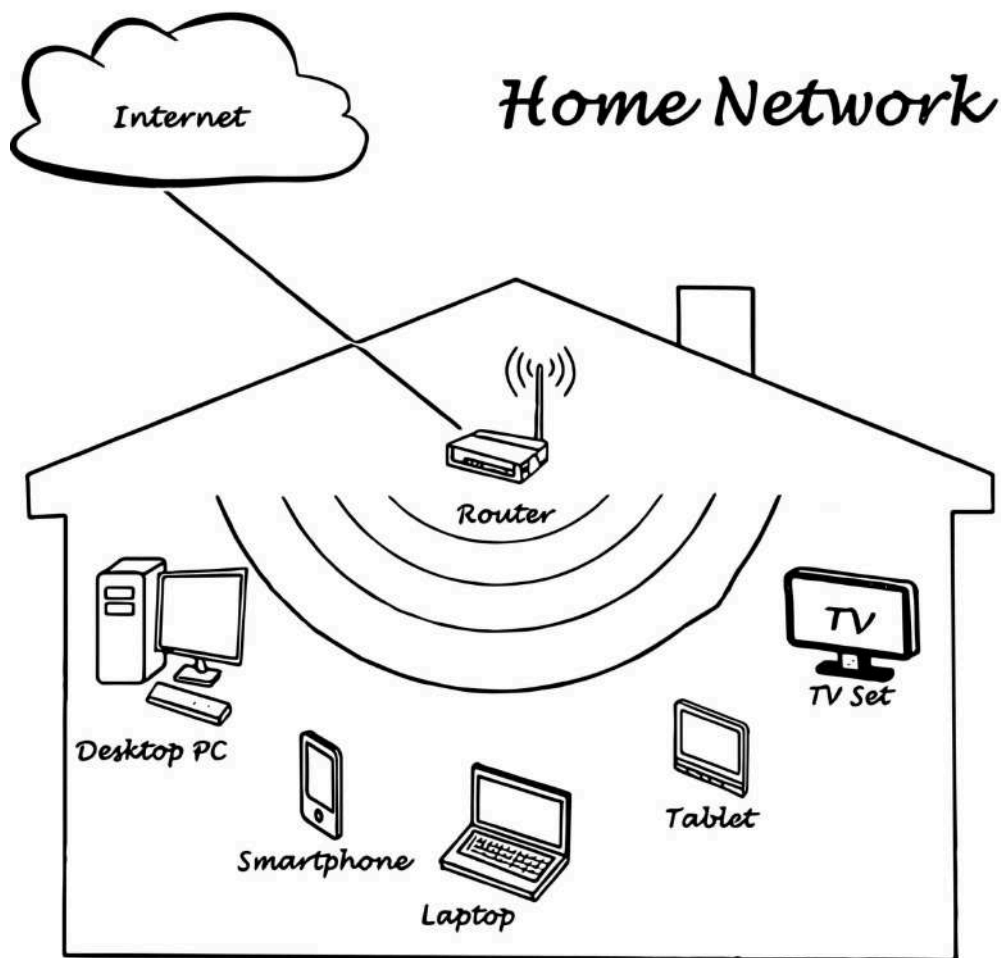
- Every computer works alone.
- File sharing requires USB drives.
- Each computer needs its own printer.
- Internet cannot be shared.

With networking:

- Share files
 - Share printers
 - Share Internet
 - Share storage
 - Share applications
 - Communicate instantly
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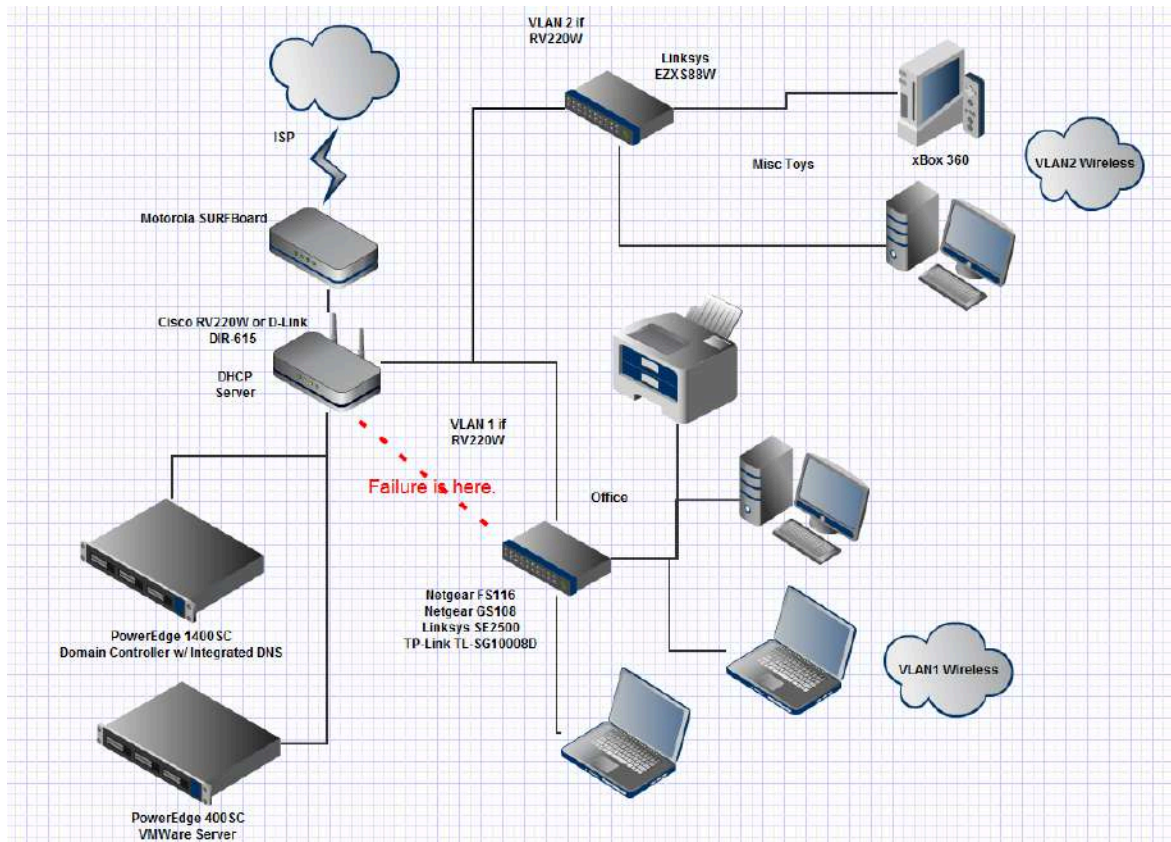
Everyday Examples

Home Network



Everyone shares the same Internet connection.

Office Network



Employees can:

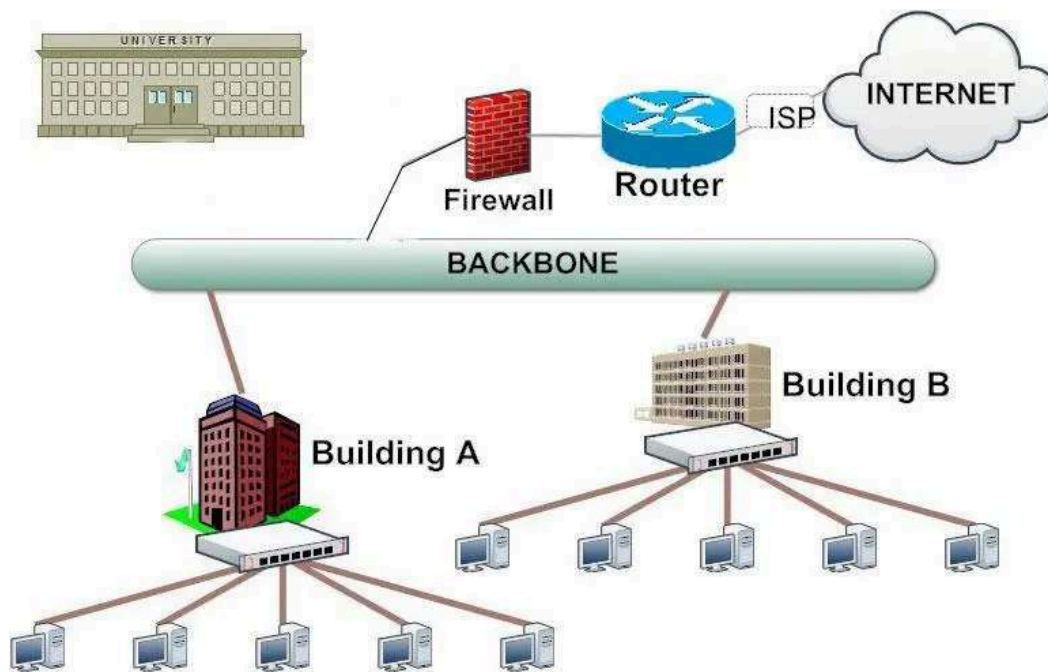
- Access company files
 - Print documents
 - Use shared software
 - Access the Internet
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University Network

Thousands of devices connect together.

Examples:

- Student computers
- Teacher laptops
- Library computers
- Wi-Fi
- CCTV
- Servers

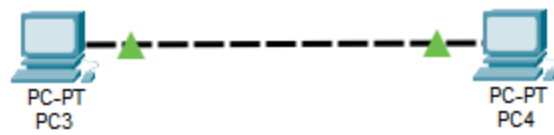


All communicate through one large network.

Benefits of Computer Networks

1. File Sharing

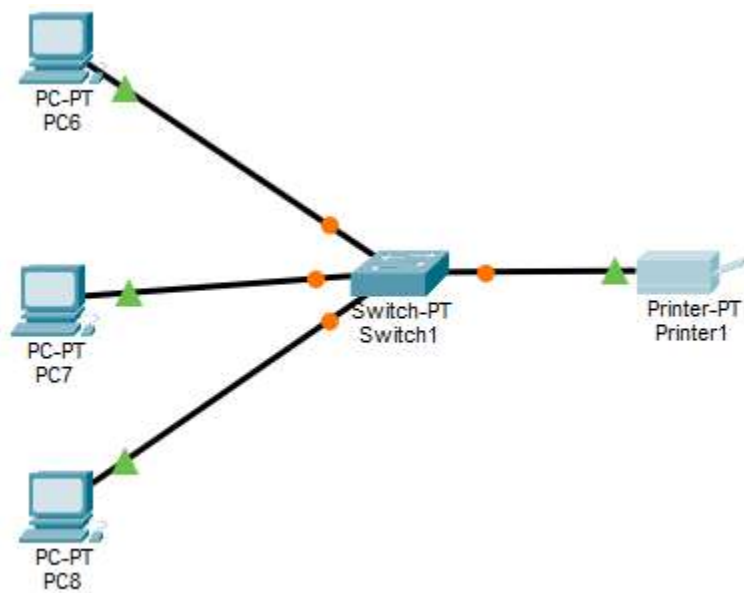
Instead of using a USB drive:



Files are transferred over the network.

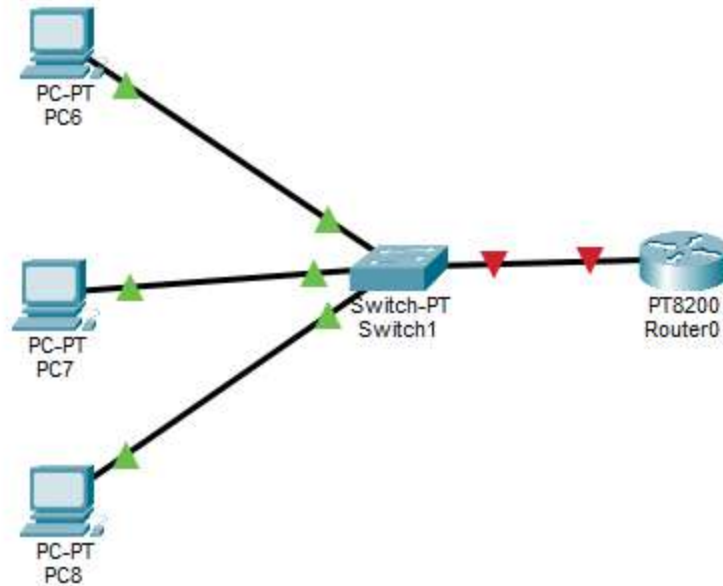
2. Printer Sharing

Instead of buying three printers:



Everyone uses one printer.

3. Internet Sharing



One Internet connection serves multiple devices.

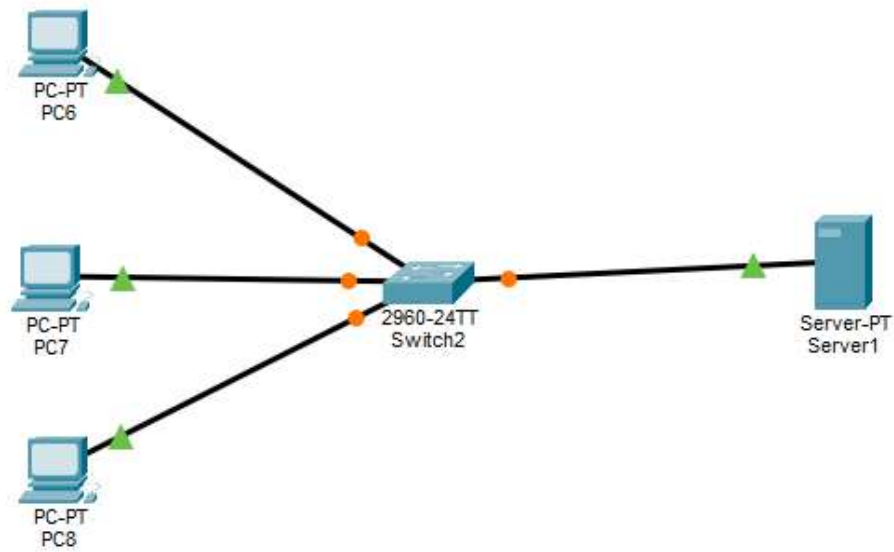
4. Communication

Examples:

- Email
- WhatsApp
- Zoom
- Microsoft Teams
- Discord

All rely on computer networks.

5. Centralized Storage



Everyone stores files on one server.

Benefits:

- Easy backup
- Better security
- Easy management

Components of a Computer Network

Component	Purpose
Computer	Sends and receives data
Server	Provides services and stores data
Switch	Connects devices within a LAN
Router	Connects different networks (e.g., your LAN to the Internet)

Access Point	Provides Wi-Fi access
Network Cable	Carries data between devices
Network Interface Card (NIC)	Allows a device to connect to a network

How Does Data Travel?

- Suppose PC1 sends a file to PC2
 - The switch forwards the data to the correct destination.
 - If the destination is on another network:
 - The router forwards the data between networks.
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Basic Networking Terms

Node

Any device connected to a network.

Examples:

- Computer
 - Printer
 - Router
 - Switch
 - Server
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Link

The connection between two devices.

Examples:

- Ethernet cable
 - Fiber optic cable
 - Wi-Fi connection
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Bandwidth

The **maximum amount of data** that can be transmitted over a network in a given time.

Example:

- 100 Mbps
- 1 Gbps
- 10 Gbps

Higher bandwidth means more data can be carried each second.

Latency

The **time it takes for data to travel** from the source to the destination.

Lower latency is better, especially for:

- Online gaming
 - Video calls
 - Real-time applications
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Packet

Data is broken into **small pieces called packets** before being sent over the network.

Instead of sending one huge file at once:

File



- Packet 1
- Packet 2
- Packet 3
- Packet 4



Destination



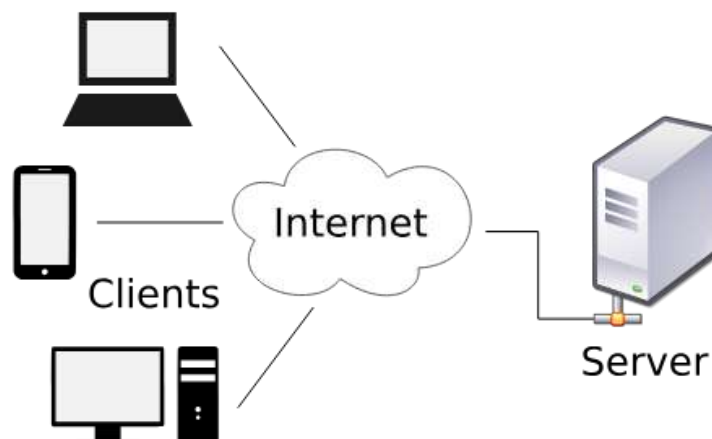
Original File

This makes transmission more reliable and efficient.

Client and Server

A **client** requests a service, and a **server** provides it.

Example:



Examples:

- A web browser requests a web page from a web server.
 - A phone requests a video from a streaming server.
 - A PC requests a file from a file server.
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Common Network Services

- Web browsing (HTTP/HTTPS)
 - Email
 - File sharing
 - Cloud storage
 - Video conferencing
 - Online gaming
 - Remote access
 - Printing
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Summary

- A **computer network** connects devices so they can communicate and share resources.
 - Devices communicate using **wired or wireless** connections.
 - **Switches** connect devices within the same network.
 - **Routers** connect different networks, such as your home network to the Internet.
 - Data is transmitted in **packets**.
 - Networks enable file sharing, Internet sharing, communication, centralized storage, and resource sharing.
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